TEXORA · LECTURE SHEET

Pattern Calculation

Turning body measurements into cuttable pattern dimensions: wearing ease, seam and hem allowances, shrinkage compensation, and the arithmetic of grading a base size into a size set.

AM-3207 · ADVANCED MACHINERIES AND RELEVANT CALCULATIONS · TOPIC 2 · SPECIALIZATION — APPAREL MANUFACTURING

1. The three additions every pattern dimension carries

A pattern piece is not the size of the body, and not even the size of the finished garment. Between the body and the cut fabric stand three calculated additions. **Wearing (and design) ease** is added to body measurements to give the *garment* dimensions — room to move, sit and breathe, plus the style's silhouette. **Seam and hem allowances** are added outside the garment's sewing lines so the pieces can be joined and finished. **Shrinkage allowance** enlarges the pattern so that after the fabric's expected contraction in washing and finishing, the garment still measures to specification. Every pattern calculation in production is one of these three additions, applied in the right order: body → garment (ease) → pattern (allowances), with shrinkage compensation applied to the fabric-dependent dimensions.

2. Worked example — the front panel of a woven shirt

The running example of this course is a men's woven shirt, base size M, body chest 100 cm. The buyer's specification gives 12 cm of total ease at the chest for this style.

WORKED EXAMPLE — VERIFY EVERY NUMBER

Garment chest = body chest + ease = 100 + 12 = **112 cm**.

Front panel sewing width (one front, quarter of the garment) = 112 ÷ 4 = **28.0 cm**.

Front panel cut width = sewing width + side seam allowance 1.5 cm = **29.5 cm** (the centre-front edge carries its own button-stand extension, drawn separately).

Body length, finished = 74 cm. The fabric shrinks 3% in length in finishing, so the pattern must be cut longer: pattern length = 74 ÷ (1 − 0.03) = 74 ÷ 0.97 = **76.3 cm** — cut at 76.3 cm, the garment finishes at 74 cm.

Armhole depth (a standard drafting rule for this shirt block) = chest ÷ 4 + 5 = 28 + 5 = **33 cm**.

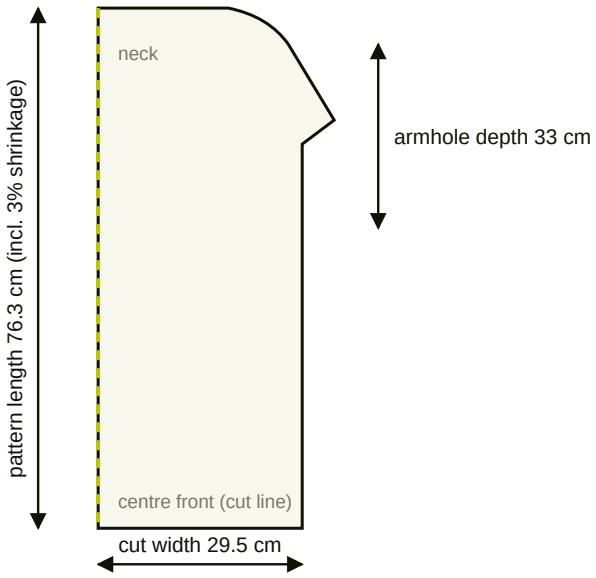


Figure 1 — Schematic front pattern block (not to scale) carrying the three calculated dimensions of the worked example. The dashed line is the centre-front cut edge; the armhole is the curved cut from shoulder to side seam.

3. Seam and hem allowances — the standard values

LOCATION	TYPICAL ALLOWANCE	REASON
Side seams, shoulder, armhole (woven shirt)	1.0–1.5 cm	Room for the seam, fray margin and later alteration
Centre back / yoke seams	1.0 cm	Flat, enclosed seams need less
Hem, woven shirt	2.0–3.0 cm total fold	A turned-up hem that lies flat and covers the raw edge
Cuff and collar edges (enclosed)	0.7–1.0 cm	Turned and closed seams; excess bulk is trimmed
Knit garments (overlocked seams)	0.7–1.0 cm	The overlock stitch itself occupies the allowance

Allowances are part of the pattern calculation, not an afterthought: the marker and fabric calculations of Topics 3 and 5 consume the *cut* dimensions, allowances included.

4. Grading calculation

Production patterns are drafted in one base size and **graded** — enlarged or reduced by fixed increments (the **grade rules**) at each point of measure. Grading is pure arithmetic: each size differs from its

neighbour by the same increment, so a four-size set is a table, not four drafting exercises. For the example shirt the grade rules are: chest 5 cm, body length 2 cm, sleeve length 1.5 cm per size step.

POINT OF MEASURE (GARMENT)	S	M (BASE)	L	XL	GRADE STEP
Chest (cm)	107	112	117	122	5.0
Front panel sewing width, chest ÷ 4 (cm)	26.75	28.0	29.25	30.5	1.25
Body length (cm)	72	74	76	78	2.0
Sleeve length (cm)	60.5	62	63.5	65	1.5

Note how a full-circumference grade of 5 cm becomes a panel grade of 5 ÷ 4 = 1.25 cm on the quarter pattern — the commonest grading error in a manual pattern room is applying the full increment to a quarter panel. CAD grading (Topic 1) applies the rule to every pattern point automatically, which is precisely why the rule, not the drawing, must be right. Tolerance is part of the calculation too: the measurement chart states each point with its tolerance (chest ± 1.5 cm is used in Topic 4 of AM-4101), and inspection judges against the graded size's own value, not the base size.

Glossary

TERM	MEANING
Ease	The difference between garment and body measurement at the same point; wearing ease for function, design ease for silhouette.
Seam allowance	Fabric between the sewing line and the cut edge, added around pattern pieces for joining and finishing.
Shrinkage allowance	Enlargement of pattern dimensions to compensate for fabric contraction, so the finished garment meets specification.
Grade rule	The fixed increment by which a point of measure changes between adjacent sizes.
Base size	The size in which the pattern is drafted and from which all other sizes are graded.

Key points and summary

- Pattern dimension = body + ease (garment) + seam/hem allowance (cut), with shrinkage compensation applied as finished ÷ (1 − shrinkage).

- Worked standard: chest 100 + 12 ease = 112 cm garment; front panel 28.0 cm sewing width, 29.5 cm cut width; length 74 cm finished needs a 76.3 cm pattern at 3% shrinkage.
- Grading applies fixed increments per size; a circumference increment divides by four on a quarter panel.
- Allowances feed directly into marker and fabric calculations — pattern arithmetic errors become fabric money errors.

Practice

Answer in your own words; brief model answers follow.

1. A trouser fabric shrinks 2% in length. The finished outseam must be 104 cm (the M-size trouser of the course measurement chart). What pattern length is cut?
2. For the example shirt, calculate the L-size front panel cut width at the chest, using the grading table and a 1.5 cm side seam allowance.
3. A pattern maker applies the full 5 cm chest grade to the quarter front panel instead of 1.25 cm. By how much is the XL garment chest wrong?
4. Why is shrinkage compensation calculated as finished ÷ (1 − s) rather than finished × (1 + s)? Show the difference for the 74 cm length at 3%.

Answer key

1. 104 ÷ 0.98 = 106.1 cm.
2. L sewing width = 117 ÷ 4 = 29.25 cm; cut width = 29.25 + 1.5 = 30.75 cm.
3. The panel gains an extra 3.75 cm per quarter, so the garment chest is 4 × 3.75 = 15 cm too large.
4. The cut length shrinks by s of the *cut* length, so finished = cut × (1 − s) and cut = finished ÷ (1 − s). At 3%: 74 ÷ 0.97 = 76.29 cm, whereas 74 × 1.03 = 76.22 cm — a small difference here, but it grows with shrinkage and is systematically short.